

## V9.03 Multi-Chain Report

Independent chains = initial structures x seeds

Structures: ['compact\_aligned', 'compact\_tilted', 'expanded\_tilted']

Seeds: [1, 2, 3, 4]

Cycles / warm-up: 2000 / 500

Magnetic sweeps per cycle: 5

Elapsed seconds: 4750.984

Sampler: Metropolis-Hastings with symmetric proposals

Diagnostics use post-warm-up (chain, draw) arrays.

Rank R-hat includes split chains and folded rank normalization.

ESS measures independent-sample information, separately for bulk/tails/mean.

MCSE(mean) is sampling uncertainty, in the variable's original units.

Failed or undefined diagnostics do not support equilibrium estimates.

Passing checks is not proof of convergence or physical model validity.

Per-start diagnostics and full settings are available in CSV/JSON.

# Per-chain post-warm-up averages

| chain | start           | seed | E_mean   | body_tilt | SL_PCA  |
|-------|-----------------|------|----------|-----------|---------|
| 0     | compact_aligned | 1    | -637.874 | 3.09678   | 3.18418 |
| 1     | compact_aligned | 2    | -637.722 | 4.71369   | 5.03165 |
| 2     | compact_aligned | 3    | -637.247 | 5.30397   | 4.97273 |
| 3     | compact_aligned | 4    | -638.428 | 6.36182   | 6.32319 |
| 4     | compact_tilted  | 1    | -636.866 | 6.72672   | 6.49351 |
| 5     | compact_tilted  | 2    | -638.033 | 3.04436   | 3.06117 |
| 6     | compact_tilted  | 3    | -637.287 | 7.27419   | 6.89041 |
| 7     | compact_tilted  | 4    | -636.973 | 4.08471   | 4.33681 |
| 8     | expanded_tilted | 1    | -601.579 | 11.4245   | 11.4327 |
| 9     | expanded_tilted | 2    | -598.655 | 11.3167   | 11.889  |
| 10    | expanded_tilted | 3    | -599.984 | 13.445    | 12.9165 |
| 11    | expanded_tilted | 4    | -592.062 | 16.2884   | 17.3855 |

# Pooled rank R-hat, ESS and MCSE

Chains: 12. Retained draws per chain: 1500.

Minimum retained draws for estimation: 100.

N/A means not estimable. It does not indicate a converged or zero-error result.

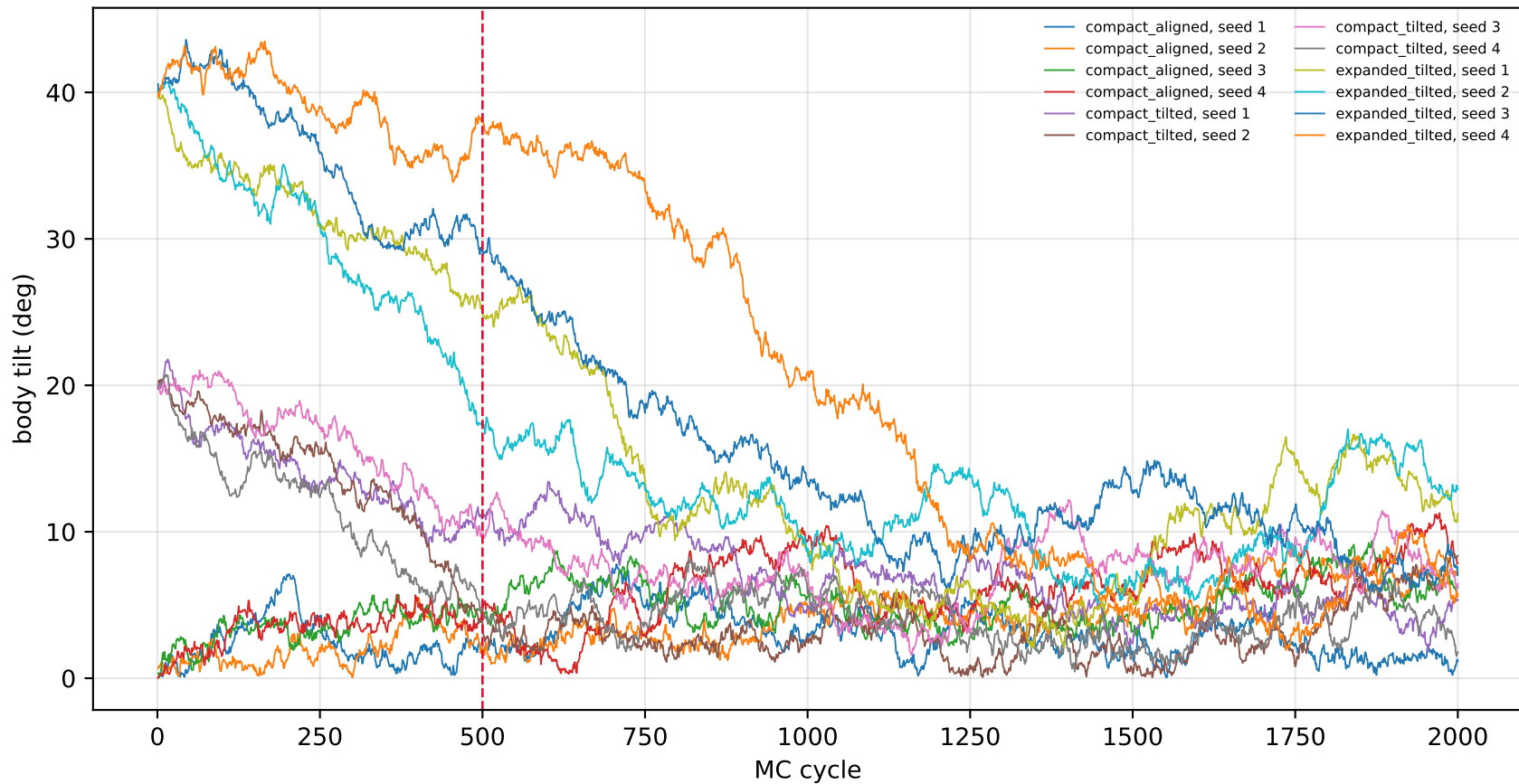
Diagnostic flags: high\_rhat,low\_ess, high\_rhat,low\_ess,high\_mcse

| variable               | rhat_rank | ess_bulk | ess_tail | ess_mean | mcse_mean   |
|------------------------|-----------|----------|----------|----------|-------------|
| energy_kBT             | 1.746     | 18.0755  | 30.044   | 13.5372  | 5.48856     |
| local_beta_deg         | 1.32264   | 28.743   | 31.4022  | 22.5487  | 0.667235    |
| body_tilt_deg          | 1.82984   | 17.4461  | 26.6883  | 16.904   | 1.50129     |
| sl_pca_tilt_deg        | 1.80712   | 17.6544  | 26.0588  | 16.4255  | 1.56055     |
| body_sl_difference_deg | 1.45996   | 23.8943  | 43.4104  | 23.3508  | 0.196592    |
| body_sl_axis_angle_deg | 1.59626   | 20.3498  | 43.7769  | 18.877   | 0.209023    |
| magnetization          | 1.27728   | 31.8868  | 36.8703  | 26.6427  | 0.0029302   |
| abs_muB                | 1.27728   | 31.8868  | 36.8703  | 26.6427  | 0.0029302   |
| body_order             | 2.73344   | 13.9764  | 20.7482  | 12.6183  | 0.00314553  |
| sl_pca_order           | 2.96382   | 13.6708  | 15.2265  | 13.3364  | 0.000974383 |
| min_gap_nm             | 1.19331   | 41.8246  | 134.832  | 43.3185  | 0.00289272  |
| rg_nm                  | 2.80168   | 13.8954  | 25.436   | 12.0754  | 0.298099    |

# Diagnostic flags

| variable               | status                      |
|------------------------|-----------------------------|
| energy_kBT             | high_rhat,low_ess           |
| local_beta_deg         | high_rhat,low_ess           |
| body_tilt_deg          | high_rhat,low_ess,high_mcse |
| sl_pca_tilt_deg        | high_rhat,low_ess,high_mcse |
| body_sl_difference_deg | high_rhat,low_ess           |
| body_sl_axis_angle_deg | high_rhat,low_ess           |
| magnetization          | high_rhat,low_ess           |
| abs_muB                | high_rhat,low_ess           |
| body_order             | high_rhat,low_ess           |
| sl_pca_order           | high_rhat,low_ess           |
| min_gap_nm             | high_rhat,low_ess           |
| rg_nm                  | high_rhat,low_ess           |

# Coherent cube-body tilt



# Superlattice PCA tilt

